ON-BOARD DIAGNOSTIC SYSTEM TEST MODE [SKYACTIV-D 1.5]

id0102q2142500

Diagnostic Test Mode

- To match the OBD regulations, the following diagnostic test modes have been supported.

Diagnostic test mode	Item
Mode 01	Sending diagnostic data (PID data monitor/On-board system readiness test)
Mode 02	Sending freeze frame data
Mode 03	Sending emission-related malfunction code (Diagnostic trouble code: DTC)
Mode 04	Clearing/resetting emission-related malfunction information
Mode 07	Sending continuous monitoring system test results (Pending code)
Mode 08	On-board device control (Simulation test, active command mode)
Mode 09	Request vehicle information

Sending Diagnostic Data (Mode 01)

PID data monitor

- The PID data monitor items are shown below.

PID data monitor table

—: Not applicable

Full names	Unit
Monitor status since DTCs cleared	—
	Undefined/Not Used
LOAD	%
ECT	°C, °F
MAP	kPa, Bar, psi
Engine speed	rpm
Vehicle speed	km/h, mph
Intake shutter valve	%
OBD requirement according to vehicle design	—
Distance travelled while check engine light is activated	km, miles
Common rail pressure	kPa, Bar, psi
Distance travelled since DTCs cleared	km, miles
Barometric pressure	kPa, Bar, psi
PCM power supply voltage	V
Relative intake shutter valve position	%
Ambient temperature	°C, °F
Accelerator pedal position No.1	%
Accelerator pedal position No.2	%
Intake shutter valve control desired value	%
Engine oil temperature	°C, °F
MAF sensor	—
	g/s
ECT	—
	°C, °F
IAT sensor No.1	—
	°C, °F
HP-EGR control valve actual opening angle	—
	%
Common rail pressure control system desired value	—
	kPa, Bar, psi
	°C, °F
MAP	—
	kPa, Bar, psi
MAP desired value	—
	kPa, Bar, psi
Variable geometry turbocharger desired position	%
Variable geometry turbocharger actual position	%
Exhaust gas pressure	—
	kPa, Bar, psi
Turbocharger turbine revolution	—
	rpm

Full names	Unit
Boost air temperature	—
	°C, °F
Exhaust gas temperature (No.1)	—
	°C, °F
Exhaust gas temperature (No.2)	—
	°C, °F
Exhaust gas temperature (No.3)	—
	°C, °F
Exhaust gas temperature (No.4)	—
	°C, °F
Diesel particulate filter	—
	kPa, Bar, psi

Sending Freeze Frame Data (Mode 02)

Freeze frame data

- The freeze frame data consists of data for vehicle and engine control system operation conditions when malfunctions in the engine control system are detected and stored in the PCM.
- Freeze frame data is stored at the instant the check engine light illuminates, and only a part of the DTC data is stored.
- For the freeze frame data, if there are several malfunctions in the engine control system, the data for the malfunction which occurred initially is stored. Thereafter, if a misfire or fuel injection control malfunction occurs, data from the misfire or fuel injection control malfunction is written over the initially stored data. However, if the initially stored freeze frame data is a misfire or fuel injection control malfunction, it is not overwritten.

Snapshot data

- The snapshot data stores the currently detected DTC data.
- The recording timing for the freeze frame data/snapshot data differs depending on the number of DTC drive cycles.
  - — For a DTC with a drive cycle number 1, only the malfunction determination data is recorded.
  - — For a DTC with a drive cycle number 2, both the malfunction determination and undetermined data is recorded.

Freeze frame data table

—: Not applicable

Freeze frame data item	Description	Unit	Corresponding PID data monitor item
LOAD	Engine load	%	LOAD
ECT	Engine coolant temperature	°C, °F	ECT
MAP	Manifold absolute pressure	KPa {MPa}, mBar {Bar}, psi, in H20	—
VS	Vehicle speed	KPH, MPH	VSS
TP	Intake shutter valve position	%	ISV_POS
FRP	Fuel pressure at common rail	KPa {MPa}, mBar {Bar}, psi, in H20	FRP
CLRDIST	Mileage after DTC cleared	km, ft, mi	CLR_DIST
BARO	Barometric pressure	KPa {MPa}, mBar {Bar}, psi, in H20	BARO
VPWR	Module supply voltage	V	VPWR
TP_REL	Intake shutter valve opening angle (relative value)	%	—
AAT	Ambient air temperature	°C, °F	IAT
APP_D	Accelerator pedal position No.1	%	APP1
APP_E	Accelerator pedal position No.2	%	APP2
TAC_PCT	Intake shutter valve control desired value	%	ISV_DSD
FUEL_TYP	Fuel type	—	—
EOT	Engine oil temperature	°C, °F	EOT
MAF_A	Mass air flow	g/sec	MAF
ECT1	Engine coolant temperature	°C, °F	—
IAT11	Intake air temperature	°C, °F	—
EGR_A_CMD	HP-EGR control valve desired value	%	HPEGR_DSD
EGR_A_ACT	HP-EGR control valve actual value	%	HPEGR_ACT

Freeze frame data item	Description	Unit	Corresponding PID data monitor item
EGR_B_ERR	Commanded EGR and EGR Error	%	—
FRP_A_CMD	Commanded fuel pressure at common rail	KPa {MPa}, mBar {Bar}, psi, in H20	FRP_DSD
FRT_A	Fuel temperature at common rail	°C, °F	FRT
BP_A_CMD	Commanded boost air pressure	KPa {MPa}, mBar {Bar}, psi, in H20	MAP_DSD
BP_A_ACT	Actual measured boost air pressure	KPa {MPa}, mBar {Bar}, psi, in H20	MAP
BP_A_ST	Boost air pressure status	Open Loop/Closed Loop/ Fault present	—
VGT_A_CMD	Turbocharger with variable turbine geometry position desired value	%	VGTURB_DSD
VGT_A_ACT	Turbocharger with variable turbine geometry position actual value	%	VGTURB_ACT
EP_1	Exhaust gas pressure sensor No.1 value	KPa {MPa}, mBar {Bar}, psi, in H20	EXHPRES1
TCA_RPM	Turbocharger with variable turbine geometry turbine speed	RPM	VGTURB_RPM
EGT11	Exhaust gas temperature sensor No.1 value	°C, °F	EXHTEMP1
EGT12	Exhaust gas temperature sensor No.2 value	°C, °F	EXHTEMP2
EGT13	Exhaust gas temperature sensor No.3 value	°C, °F	EXHTEMP3
EGT14	Exhaust gas temperature sensor No.4 value	°C, °F	EXHTEMP4
DPF1_DP	Exhaust gas pressure sensor No.2 value	KPa {MPa}, mBar {Bar}, psi, in H20	EXHPRESS_DIF

Snapshot data table

—: Not applicable

Snapshot data item	Definition	Data read/use method	Unit	Corresponding PID data monitor item
ALTT_V	Generator output voltage	—	V	ALTT V
MIL	Check engine light illumination status	—	Off/On	MIL
APP1	Accelerator pedal position No.1 voltage	—	V	APP1
	Accelerator pedal position No.1		%	
APP2	Accelerator pedal position No.2 voltage	—	V	APP2
	Accelerator pedal position No.2		%	
ALT_CUR_DSD	Generator target generated current	—	A	—
BATT_RES	Battery internal resistance (estimated)	—	ohm	BATT_RES
TOTAL_TIME	Total energization time for module	The elapsed time when the start stop unit detected a DTC can be calculated by performing the following procedure. 1. Verify the PID item TOTAL_TIME of the instrument cluster. 2. Verify the snapshot data item TOTAL_TIME. 3. Subtract 2 from 1.	hh:mm:ss	—

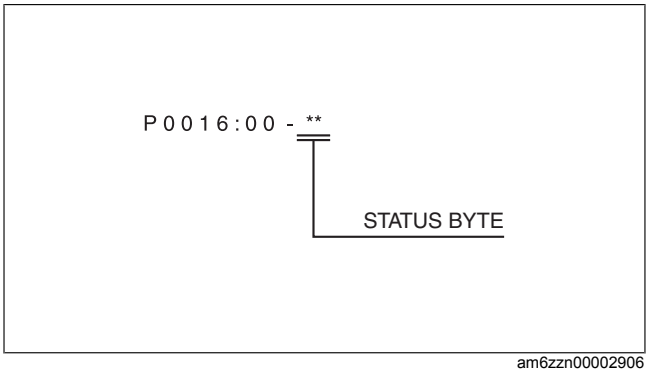
Snapshot data item	Definition	Data read/use method	Unit	Corresponding PID data monitor item
TOTAL_DIST	Total distance	The distance traveled when the start stop unit detected a DTC can be calculated by performing the following procedure. 1. Verify the odometer value in the instrument cluster. 2. Verify the snapshot data item TOTAL_DIST. 3. Subtract 2 from 1.	km, ft, mi	—
FUELSYS	Fuel system status	—	OL/CL/OL-Drive/OL-Fault/CL-Fault	—
LOAD_C	Calculated engine load	—	%	LOAD
ECT	Engine coolant temperature	—	°C, °F	ECT
RPM	Engine speed	—	RPM	RPM
VSS	Vehicle speed	—	KPH, MPH	VSS
IAT	Intake air temperature No.1	—	°C, °F	—
MAF	Mass airflow	—	g/sec	—
TP1	Intake shutter valve position	—	%	ISV_POS
EG_RUN_TIME	Time from engine start	—	hh:mm:ss	—
FUEL_PRES	Fuel pressure	—	KPa {MPa}, mBar {Bar}, psi, in H20	FRP
SEGRP_DSD	Target EGR valve position	—	%	—
EVAPCP	Purge solenoid valve controlled value	—	%	—
FLI	Fuel level in fuel tank	—	%	—
CLR_CNT	Number of warm-up cycle after DTC cleared	—	—	—
CLR_DIST	Mileage after DTC cleared	—	km, ft, mi	CLR_DIST
FTP	Fuel tank pressure	—	KPa {MPa}, mBar {Bar}, psi, in H20	—
BARO	Barometric pressure	—	KPa {MPa}, mBar {Bar}, psi, in H20	BARO
CATT11_DSD	Estimated catalytic converter temperature	—	°C, °F	—
VPWR	Module supply voltage	—	V	VPWR
EQ_RAT11_DSD	Target equivalence ratio (lambda)	—	—	—
TP_REL	Relative throttle position	—	%	—
AAT	Ambient air temperature	—	°C, °F	IAT
TP2	Throttle valve position No.2	—	%	—
ETC_DSD	Intake shutter valve control desired value	—	%	ISV_DSD
FUEL_TYP	Type of fuel currently being utilized by the vehicle	—	NONE/Gas/METH/ETH/DSL/ELEC	—
SHRTFT12	Short term fuel trim (HO2S)	—	%	—
LONGFT12	Long term fuel trim (HO2S)	—	%	—
FRP	Fuel pressure at common rail (absolute)	—	KPa {MPa}, mBar {Bar}, psi, in H20	—
APP_R	Relative accelerator pedal position	—	%	—
EOT	Engine oil temperature	—	°C, °F	EOT
FUEL_TIMING	Main fuel injection timing related to top dead center	—	°(deg)	—
FUEL_RATE	Percentage of fuel consumption	—	l/min	—
TQ_DD	Engine torque requested by driver	—	%	—
TQ_ACT	Actual engine torque value	—	%	—

Snapshot data item	Definition	Data read/use method	Unit	Corresponding PID data monitor item
MAF_A	MAF sensor value	—	g/sec	MAF
ECT1_SUP	Engine coolant temperature No.1 support	—	No/Yes	—
ECT2_SUP	Engine coolant temperature No.2 support	—	No/Yes	—
ECT1	Engine coolant temperature No.1	—	°C, °F	—
ECT2	Engine coolant temperature No.2	—	°C, °F	—
IAT11	Intake air temperature No.1 value	—	°C, °F	—
IAT12	Intake air temperature No.2 value	—	°C, °F	—
IAT13	Intake air temperature No.3 value	—	°C, °F	—
EGRP	EGR valve position	—	%	HPEGR_DSD
EGRP_ACT	EGR valve actual opening angle	—	%	HPEGR_ACT
EGR_ERR	Percentage of abnormal EGR	—	%	—
LPEGRP	Low pressure EGR valve position	—	%	—
LPEGRP_ACT	Low pressure EGR valve actual opening angle	—	%	—
LPEGR_ERR	Percentage of abnormal low pressure side EGR	—	%	—
EGRT11	Exhaust gas recirculation temperature	—	°C, °F	—
EGRT12	Low pressure side exhaust gas recirculation temperature	—	°C, °F	—
TAC_A_CMD	Commanded throttle valve actuator control	—	%	—
TP_A_REL	Relative throttle valve position	—	%	—
FRP_A_CMD	Commanded fuel pressure at common rail	—	KPa {MPa}, mBar {Bar}, psi, in H20	FRP_DSD
FRP_A	Fuel pressure at common rail	—	KPa {MPa}, mBar {Bar}, psi, in H20	—
FRT	Fuel temperature at common rail	—	°C, °F	FRT
TCA_CINP	Manifold absolute pressure No.1 value	—	KPa {MPa}, mBar {Bar}, psi, in H20	—
TCB_CINP	Manifold absolute pressure No.1 value	—	KPa {MPa}, mBar {Bar}, psi, in H20	—
BP_A_CMD	Commanded manifold absolute pressure No.2	—	KPa {MPa}, mBar {Bar}, psi, in H20	MAP_DSD
BP_A_ACT	Manifold absolute pressure No.2 value	—	KPa {MPa}, mBar {Bar}, psi, in H20	MAP
BP_A_ST	Manifold absolute pressure No.2 control status	—	Open Loop/ Closed Loop/ Fault present	—
VGT_A_CMD	Turbocharger with variable turbine geometry position desired value	—	%	VGTURB_DSD
VGT_A_ACT	Turbocharger with variable turbine geometry position actual value	—	%	VGTURB_ACT
REGVP_DSD	Regulating valve position desired value	—	%	—
REGRP	Regulating valve position	—	%	—
EP_1	Exhaust gas pressure sensor No.1 value	—	KPa {MPa}, mBar {Bar}, psi, in H20	EXHPRES1
TCA_RPM	Turbocharger with variable turbine geometry turbine speed	—	RPM	VGTURB_RPM
CACT11	Boost air temperature sensor No.1 value	—	°C, °F	CACT11

Snapshot data item	Definition	Data read/use method	Unit	Corresponding PID data monitor item
EXHTEMP1	Exhaust gas temperature sensor No.1 value	—	°C, °F	EXHTEMP1
EXHTEMP2	Exhaust gas temperature sensor No.2 value	—	°C, °F	EXHTEMP2
EXHTEMP3	Exhaust gas temperature sensor No.3 value	—	°C, °F	EXHTEMP3
EXHPRESS_DIF	Exhaust gas pressure sensor No.2 value	—	KPa {MPa}, mBar {Bar}, psi, in H20	EXHPRESS_DIF
NOX_ADS_DESULF	Desulfuration condition of NOx adsorbent	—	No/Yes	—
NOX_ADS_REGEN	Regeneration condition of NOx adsorbent	—	No/Yes	—
DPF_REGN_TYP	Diesel particulate filter regeneration type	—	Passive/Active	—
DPF_REGN_STAT	Diesel particulate filter regeneration condition	—	No/Yes	—
DPF_REGEN_PCT	Specified value for performing diesel particulate filter regeneration	—	%	—
DPF_REGN_AVGD	Average of drive distance from previous diesel particulate filter regeneration to next diesel particulate filter regeneration	—	km, ft, mi	—

Status Byte for DTC

- The two digits (two digits after hyphen (-)) after the 7-digit DTC.
- A code which indicates the pending code, current/past malfunction status, or warning illumination status.
- Can be read by performing a CMDTC self-test using the Mazda Modular Diagnostic System (M-MDS).
- For details on the status byte, refer to the explanation on the Mazda Modular Diagnostic System (M-MDS) when reading the DTC.



am6zzn00002906

Sending Emission-related Malfunction Code (DTC) (Mode 03)

- The DTCs are shown below.

×: Applicable
—: Not applicable

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
B10A2:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Vehicle collision	—	1	Other	C	—
P0016:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Camshaft position/ Crankshaft position correlation problem	×	1	CCM	C, R	×

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P0030:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	A/F sensor heater control circuit range/ performance problem	×	1	A/F sensor	C	×
P0047:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Regulating solenoid valve control circuit low input	×	1	Turbo system	C, O, R	×
P0048:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Regulating solenoid valve control circuit high input	×	1	Turbo system	C, O, R	×
P0072:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Ambient temperature sensor circuit low input	—	1	Other	C, O, R	—
P0073:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Ambient temperature sensor circuit high input	—	1	Other	C, O, R	—
P007B:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Boost air temperature sensor circuit range/ performance problem	×	2	Turbo system	C	×
P007C:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Boost air temperature sensor circuit low input	×	1	Turbo system	C, O, R	×
P007D:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Boost air temperature sensor circuit high input	×	1	Turbo system	C, O, R	×
P0087:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Low pressure malfunction in common rail fuel pressure control system	×	1	Fuel system	C	×
P0088:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	High pressure malfunction in common rail fuel pressure control system	×	1	Fuel system	C	×
P0089:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	High pressure malfunction in common rail fuel pressure control system	×	1	Fuel system	C	×
P0093:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel leakage from fuel pressure control system	×	1	Fuel system	C	×
P0096:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	IAT sensor No.2 circuit range/ performance problem	×	2	Fuel system	C	×

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P0097:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	IAT sensor No.2 circuit low input	×	1	Fuel system	C, O, R	×
P0098:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	IAT sensor No.2 circuit high input	×	1	Fuel system	C, O, R	×
P009B:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel pressure relief valve control circuit open	×	1	Fuel system	C	×
P009C:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel pressure relief valve control circuit low input	×	1	Fuel system	C, R	×
P009D:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel pressure relief valve control circuit high input	×	1	Fuel system	C, R	×
P009E:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel pressure relief valve malfunction (stuck close)	×	1	Fuel system	C	×
P009F:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel pressure relief valve malfunction (stuck on)	×	1	Fuel system	C	×
P0101:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	MAF sensor circuit range/ performance problem	×	1	CCM	C, O	×
P0102:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	MAF sensor circuit low input	×	1	CCM	C, O, R	×
P0103:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	MAF sensor circuit high input	×	1	CCM	C, O, R	×
P0106:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	MAP sensor No. 2 circuit range/ performance problem	×	1	Turbo system	C	×
P0107:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	MAP sensor No. 2 circuit low input	×	1	Turbo system	C, O, R	×
P0108:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	MAP sensor No. 2 circuit high input	×	1	Turbo system	C, O, R	×
P0111:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	IAT sensor No.1 circuit range/ performance problem	×	2	CCM	C	×
P0112:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	IAT sensor No.1 circuit low input	×	1	CCM	C, O, R	×
P0113:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	IAT sensor No.1 circuit high input	×	1	CCM	C, O, R	×
P0116:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	ECT sensor circuit range/ performance problem	×	1	CCM	C	×
P0117:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	ECT sensor circuit low input	×	1	CCM	C, O, R	×

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P0118:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	ECT sensor circuit high input	×	1	CCM	C, O, R	×
P0122:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Intake shutter valve position sensor circuit low input	×	1	CCM	C, O, R	×
P0123:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Intake shutter valve position sensor circuit high input	×	1	CCM	C, O, R	×
P0130:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	A/F sensor circuit range/ performance problem	×	1	A/F sensor	C, O, R	×
P0131:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	A/F sensor circuit low input	×	1	A/F sensor	C, O, R	×
P0132:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	A/F sensor circuit high input	×	1	A/F sensor	C, O, R	×
P0134:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	A/F sensor circuit no activity detected	×	1	A/F sensor	C	×
P0181:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel temperature sensor circuit range/ performance problem	×	2	Fuel system	C	×
P0182:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel temperature sensor circuit low input	×	1	Fuel system	C, O, R	×
P0183:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel temperature sensor circuit high input	×	1	Fuel system	C, O, R	×
P0191:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel pressure sensor circuit range/ performance problem	×	1	Fuel system	C	×
P0192:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel pressure sensor circuit low input	×	1	Fuel system	C, O, R	×
P0193:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel pressure sensor circuit high input	×	1	Fuel system	C, O, R	×
P0196:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Engine oil temperature sensor circuit range/ performance problem	×	2	CCM	C	×
P0197:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Engine oil temperature sensor circuit low input	×	1	CCM	C, O, R	×

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P0198:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Engine oil temperature sensor circuit high input	×	1	CCM	C, O, R	×
P0201:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 1 circuit open	×	1	Fuel system	C, R	×
P0202:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 2 circuit open	×	1	Fuel system	C, R	×
P0203:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 3 circuit open	×	1	Fuel system	C, R	×
P0204:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 4 circuit open	×	1	Fuel system	C, R	×
P0219:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Engine overspeed condition	—	1	Other	C	—
P0234:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Turbocharger overboost condition	—	2	Turbo system	C	×
P026A:00	OFF	ON	OFF	OFF	OFF	OFF	OFF	Charge air cooler efficiency below threshold	×	1	Turbo system	C	×
P0299:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Turbocharger underboost condition	—	2	Turbo system	C	×
P02EE:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 1 circuit: shorted to each other	×	1	Fuel system	C, R	×
P02EF:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 2 circuit: shorted to each other	×	1	Fuel system	C, R	×
P02F0:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 3 circuit: shorted to each other	×	1	Fuel system	C, R	×
P02F1:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 4 circuit: shorted to each other	×	1	Fuel system	C, R	×
P0301:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Cylinder No.1 misfire	×	1	CCM	C	×
P0302:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Cylinder No.2 misfire	×	1	CCM	C	×
P0303:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Cylinder No.3 misfire	×	1	CCM	C	×
P0304:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Cylinder No.4 misfire	×	1	CCM	C	×
P0313:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Misfire detected with low fuel	×	1	Other	C	×
P0336:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	CKP sensor circuit range/ performance problem	×	2	CCM	C, R	×
P0337:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	CKP sensor circuit problem	×	1	CCM	C, R	×
P0339:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	CKP sensor circuit range/ performance problem	×	1	CCM	C	—

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P0341:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	CMP sensor circuit range/ performance problem	×	2	CCM	C, R	×
P0342:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	CMP sensor circuit problem	×	1	CCM	C, R	×
P0383:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Glow control module circuit low input	×	1	CCM	C	×
P0384:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Glow control module circuit high input	×	1	CCM	C	×
P0401:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	HP-EGR flow insufficient detected	×	2	EGR system	C	×
P0402:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	HP-EGR flow excessive detected	×	1	EGR system	C	×
P0404:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Circuit malfunction in HP-EGR control valve DC motor control system	×	1	EGR system	C, O, R	×
P0405:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	HP-EGR control valve position sensor circuit low input	×	1	EGR system	C, O, R	×
P0406:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	HP-EGR control valve position sensor circuit high input	×	1	EGR system	C, O, R	×
P0407:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	LP-EGR control valve position sensor circuit low input	×	1	EGR system	C, O, R	×
P0408:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	LP-EGR control valve position sensor circuit high input	×	1	EGR system	C, O, R	×
P0421:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Warm up catalyst system efficiency below threshold	—	2	Catalyst	C	×
P045B:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	LP-EGR control valve control: circuit malfunction	×	1	EGR system	C, O, R	×
P0471:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas pressure sensor No.1 circuit range/ performance problem	×	2	Turbo system	C	×
P0472:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas pressure sensor No.1 circuit low input	×	1	Turbo system	C, O, R	×

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P0473:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas pressure sensor No.1 circuit high input	×	1	Turbo system	C, O, R	×
P0475:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust shutter valve circuit problem	×	1	EGR system	C, O, R	×
P0476:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust shutter valve circuit range/ performance problem	×	1	EGR system	C	×
P0480:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Fan control module control circuit problem	—	1	Other	C, O	—
P0488:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Duty signal error in HP-EGR control valve control system	×	1	EGR system	C	×
P048D:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust shutter valve position sensor circuit low input	×	1	EGR system	C, O, R	×
P048E:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust shutter valve position sensor circuit high input	×	1	EGR system	C, O, R	×
P049B:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	LP-EGR flow insufficient detected	×	2	EGR system	C	×
P049C:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	LP-EGR flow excessive detected	×	1	EGR system	C	×
P0500:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	VSS circuit problem	×	1	CCM	C, O, R	×
P0504:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Brake switch circuit problem	×	2	CCM	C	×
P0522:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Engine oil pressure sensor circuit low input	×	1	CCM	C, O, R	×
P0523:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Engine oil pressure sensor circuit high input	×	1	CCM	C, O, R	×
P0524:00	ON	OFF	OFF	OFF	ON	OFF	OFF	Engine oil pressure too low	×	1	CCM	C	×
P0532:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Refrigerant pressure sensor circuit low input	×	1	Other	C	×
P0533:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Refrigerant pressure sensor circuit high input	×	1	Other	C	×
P053B:00	OFF	ON	OFF	OFF	OFF	OFF	OFF	Blow-by heater relay control circuit low input	—	1	Other	C, O, R	—
P053C:00	OFF	ON	OFF	OFF	OFF	OFF	OFF	Blow-by heater relay control circuit high input	—	1	Other	C, O, R	—

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P0545:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas temperature sensor No.1 circuit low input	×	1	CCM	C, O, R	×
P0546:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas temperature sensor No.1 circuit high input	×	1	CCM	C	×
P0555:00	OFF	OFF	Flash	OFF	OFF	OFF	OFF	Power brake unit vacuum sensor circuit problem	×	1	Other	C, R	—
P055F:00	ON	OFF	OFF	OFF	ON	OFF	OFF	Engine oil pressure control circuit low oil pressure	×	1	CCM	C	×
P057F:00	OFF	ON	Flash	OFF	OFF	OFF	OFF	Power system: Battery deterioration	×	1	Other	C	—
P058A:00	OFF	OFF	Flash	ON*4	OFF	OFF	OFF	Current sensor: Function malfunction	×	1	Other	C	—
P0601:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	PCM memory check sum error	×	1	CCM	C, O, R	×
P0602:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	PCM programming error	×	1	CCM	C	—
P0605:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	PCM memory check sum error	×	1	CCM	C	×
P0606:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	PCM processor error	×	1	CCM	C, O, R	×
P0607:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Control module performance problem	×	1	CCM	C, O, R	×
P0610:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	PCM vehicle configuration error	×	1	CCM	C	—
P0615:00	OFF	OFF	Flash	OFF	OFF	OFF	OFF	Starter relay circuit problem	—	1	Other	C	×
P062A:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Suction control valve circuit problem	×	1	Fuel system	C, R	×
P062B:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	PCM internal malfunction	×	1	Fuel system	C, O, R	×
P0642:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Constant voltage power supply circuit low input	×	1	CCM	C	×
P0643:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Constant voltage power supply circuit high input	×	1	CCM	C	×
P0646:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	A/C relay circuit low input	—	1	Other	C	—
P0647:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	A/C relay circuit high input	—	1	Other	C	—

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P0652:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Constant voltage power supply control circuit low input	×	1	CCM	C	×
P0653:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Constant voltage power supply control circuit high input	×	1	CCM	C	×
P0670:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Glow control module control circuit problem	—	1	CCM	C, O	×
P0671:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Glow plug No.1 control circuit problem	—	1	CCM	C, O	×
P0672:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Glow plug No.2 control circuit problem	—	1	CCM	C, O	×
P0673:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Glow plug No.3 control circuit problem	—	1	CCM	C, O	×
P0674:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Glow plug No.4 control circuit problem	—	1	CCM	C, O	×
P0698:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Constant voltage power supply circuit low input	×	1	CCM	C	×
P0699:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Constant voltage power supply circuit high input	×	1	CCM	C	×
P06B8:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Internal control module non-volatile RAM error	×	1	CCM	C	×
P06DB:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Engine oil solenoid valve circuit low input	×	1	CCM	C, O, R	—
P06DC:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Engine oil solenoid valve control circuit high input	×	1	CCM	C, O, R	—
P06DD:00	OFF	ON	OFF	OFF	OFF	OFF	OFF	Engine oil pressure switch control circuit high input	—	1	Other	C	—
P06DE:00	OFF	ON	OFF	OFF	OFF	OFF	OFF	Engine oil pressure switch control circuit low input	—	1	Other	C	—
P0703:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Brake switch input circuit problem	×	2	CCM	C	—
P0704:00 *2	ON	OFF	OFF	OFF	OFF	OFF	OFF	CPP switch input circuit problem	×	2	CCM	C	×

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P07BE:00*2	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Transmission indeterminate failure (failed to neutral)	×	1	Other	C	—
P0850:00*2	ON	OFF	OFF	OFF	OFF	OFF	OFF	Neutral switch No.1 input circuit problem	×	2	CCM	C	×
P0A06:00	OFF	ON	OFF	OFF	OFF	OFF	OFF	Electric water pump control circuit low input	—	1	Other	C, O	—
P0A07:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Electric water pump control circuit low input	—	1	Other	C, O	—
P0A0F:00	OFF	OFF	Flash	OFF	OFF	OFF	OFF	Engine failed to restart	×	1	Other	C	×
P0A8D:00	OFF	ON	Flash	OFF	OFF	OFF	OFF	Power supply system circuit low input	×	1	Other	C	—
P0A8F:00	OFF	OFF	Flash	OFF	OFF	OFF	OFF	Power system: Low input	×	1	Other	C	—
P0A94:00	OFF	OFF	Flash	OFF	OFF	OFF	OFF	DC-DC converter: control circuit signal error (Without i-ELOOP) DC-DC converter (i-ELOOP): control circuit signal error (With i-ELOOP)	×	1	Other	C	—
P0C73:00	OFF	ON	OFF	OFF	OFF	OFF	OFF	Electric water pump control: abnormal operation	—	1	Other	C	—
P1051:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Injection quantity too low cylinder No.1	×	2	Fuel system	C	×
P1052:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Injection quantity too high cylinder No.1	×	2	Fuel system	C	×
P1053:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Injection quantity too low cylinder No.2	×	2	Fuel system	C	×
P1054:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Injection quantity too high cylinder No.2	×	2	Fuel system	C	×
P1055:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Injection quantity too low cylinder No.3	×	2	Fuel system	C	×
P1056:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Injection quantity too high cylinder No.3	×	2	Fuel system	C	×
P1057:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Injection quantity too low cylinder No.4	×	2	Fuel system	C	×

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P1058:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Injection quantity too high cylinder No.4	×	2	Fuel system	C	×
P111A:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Engine coolant temperature is high	—	1	Other	C	—
P1142:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Fuel Restriction Condition	×	1	Other	C	×
P115A:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Lack of remaining fuel (output restriction)	×	1	Other	C	—
P115B:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Lack of remaining fuel (fuel injection pause)	×	1	Other	C	—
P117A:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Record of torque restriction for piston protection	—	1	Other	C	—
P1200:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injection amount learning not completed	×	1	Other	C	—
P1260:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Immobilizer system problem	—	1	Other	C	—
P1282:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Common rail control system problem	×	1	Fuel system	C	×
P1284:00	ON	ON	OFF	OFF	OFF	OFF	OFF	Fuel injector carbon deposit burn-off procedure not completed	—	1	CCM	C	×
P1329:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Common rail pressure higher than desired (engine running)	×	1	Fuel system	C	×
P1336:00	OFF	ON	OFF	OFF	OFF	OFF	OFF	CKP sensor/ CMP sensor malfunction	—	1	CCM	C	—
P1378:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector circuit low input	×	1	Fuel system	C, R	×
P1379:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector circuit high input	×	1	Fuel system	C, R	×
P1589:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Intake shutter valve control duty signal error	×	1	CCM	C	×
P1675:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector code not programmed	×	1	Fuel system	C	×
P1676:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector code mistakenly programmed	×	1	Fuel system	C	×

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P167B:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injection amount learning not completed in fuel injection system	×	1	Other	C	—
P176E:00*2	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Clutch stroke sensor/starter interlock switch correlation problem	×	1	Other	C	—
P1905:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Check connector circuit low input	—	1	Other	C	—
P2002:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Diesel particulate filter function decreased	—	2	Diesel particulate filter system	C	×
P2032:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas temperature sensor No.2 circuit low input	×	1	CCM	C, O, R	×
P2033:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas temperature sensor No.2 circuit high input	×	1	CCM	C	×
P2101:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Intake shutter valve control circuit overcurrent	×	1	CCM	C, O, R	×
P2104:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Intake shutter valve control system: Idle air control	×	1	CCM	C	×
P2105:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Intake air shutter valve actuator control system malfunction (forced engine shutdown)	—	1	Other	C	—
P2106:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Throttle Actuator Control System - Forced Limited Power	×	1	CCM	C	×
P2118:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Intake shutter valve control duty signal error	×	2	CCM	C	×
P2122:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	APP sensor No. 1 circuit low input	×	1	CCM	C, O, R	×
P2123:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	APP sensor No. 1 circuit high input	×	1	CCM	C, O, R	×
P2127:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	APP sensor No. 2 circuit low input	×	1	CCM	C, O, R	×

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P2128:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	APP sensor No. 2 circuit high input	×	1	CCM	C, O, R	×
P2138:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	APP sensor No. 1/No.2 correlation problem	×	1	CCM	C	×
P213B:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	LP-EGR valve control: high duty malfunction	×	1	EGR system	C	×
P2146:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 1 and No.4 open	×	1	Fuel system	C, R	×
P2147:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 1 and No.4 circuit low input	×	1	Fuel system	C, R	×
P2148:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 1 and No.4 circuit high input	×	1	Fuel system	C, R	×
P2149:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 2 and No.3 circuit open	×	1	Fuel system	C, R	×
P2150:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 2 and No.3 circuit low input	×	1	Fuel system	C, R	×
P2151:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Fuel injector No. 2 and No.3 circuit high input	×	1	Fuel system	C, R	×
P2227:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	BARO sensor circuit range/ performance problem	×	1	CCM	C	×
P2228:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	BARO sensor circuit low input	×	1	CCM	C, O, R	×
P2229:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	BARO sensor circuit high input	×	1	CCM	C, O, R	×
P2263:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Turbocharger solenoid valve control circuit range/ performance problem	×	2	Turbo system	C	×
P2299:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Accelerator pedal: spring back malfunction	—	1	Other	C	—
P242C:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas temperature sensor No.3 circuit low input	×	1	CCM	C, O, R	×
P242D:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas temperature sensor No.3 circuit high input	×	1	CCM	C	×

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P242F:00	ON	OFF	OFF	OFF	OFF	OFF	Flash	Diesel particulate filter excess accumulation (exceeded acceptable amount)	×	1	CCM	C	×
P243C:00	OFF	OFF	OFF	OFF	OFF	OFF	Flash	Abnormal diesel particulate filter regeneration interval short distance	—	1	Other	C	—
P244A:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas pressure sensor No.2 range/ performance problem	×	1	Diesel particulate filter system	C	×
P2452:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Signal malfunction in exhaust gas pressure sensor No.2	—	1	Diesel particulate filter system	C	—
P2454:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas pressure sensor No.2 circuit low input	×	1	Diesel particulate filter system	C, O, R	×
P2455:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas pressure sensor No.2 circuit high input	×	1	Diesel particulate filter system	C, O, R	×
P2456:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Characteristic malfunction in exhaust gas pressure sensor No.2	×	1	Diesel particulate filter system	C	×
P2458:00	OFF	OFF	OFF	OFF	OFF	OFF	Flash	Diesel particulate filter excess accumulation (mid)	×	1	Other	C	—
P2459:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Abnormal diesel particulate filter regeneration interval short distance*5	—	1	Other	C	×
P2463:00	ON	OFF	OFF	OFF	OFF	OFF	Flash	Diesel particulate filter excess accumulation (large)	×	1	CCM	C	×
P246C:00	OFF	OFF	OFF	OFF	ON	OFF	OFF	Engine oil dilution, oil pressure decreased	—	1	Other	C	—

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P2470:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas temperature sensor No.4 circuit low input	×	1	CCM	C, O, R	×
P2471:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Exhaust gas temperature sensor No.4 circuit high input	×	1	CCM	C	×
P2502:00	OFF	OFF	Flash	ON*4	OFF	OFF	OFF	Generator system: Malfunction in voltage generated by generator	×	1	Other	C	—
P2503:00	OFF	OFF	Flash	ON*4	OFF	OFF	OFF	Generator system: Voltage generated by generator is low	×	1	Other	C	—
P2504:00	OFF	OFF	Flash	ON*4	OFF	OFF	OFF	Generator system: Voltage generated by generator is high	×	1	Other	C	—
P2507:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	PCM battery voltage low input	×	1	CCM	C	×
P250A:00	OFF	ON	OFF	OFF	OFF	OFF	OFF	Engine oil level signal: engine oil level sensor malfunction	—	1	Other	C	—
P250C:00	OFF	OFF	OFF	OFF	ON	OFF	OFF	Engine oil level sensor power supply circuit malfunction	—	1	Other	C	—
P250F:00	OFF	OFF	OFF	OFF	OFF	ON	OFF	Engine oil level signal: engine oil level low	—	1	Other	C	—
P252F:00	OFF	OFF	OFF	OFF	OFF	ON	OFF	Engine oil level too high	—	1	Other	C	—
P253F:00	OFF	OFF	OFF	OFF	ON	OFF	OFF	Engine oil deteriorated	—	1	Other	C	—
P2564:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Actuator position sensor circuit low input	×	1	Turbo system	C, R	×
P2565:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Actuator position sensor circuit high input	×	1	Turbo system	C, O, R	×
P2579:00	OFF	ON	OFF	OFF	OFF	OFF	OFF	Speed sensor assy. (turbocharger) range/ performance problem	×	1	Turbo system	C	—
P2580:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Speed sensor assy. (turbocharger) circuit low input	×	1	Turbo system	C	×

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
P2581:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Speed sensor assy. (turbocharger) circuit high input	×	1	Turbo system	C	×
P2610:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Instrument cluster internal engine off timer performance problem	×	2	CCM	C	×
P2681:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Circuit malfunction in coolant control valve control system	×	1	CCM	C, O, R	×
P2682:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Coolant control valve position sensor circuit low input	—	1	CCM	C, O, R	×
P2683:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Coolant control valve position sensor circuit high input	—	1	CCM	C, O, R	×
P26A3:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Coolant control valve control: high duty malfunction	×	1	CCM	C, O	×
P2A00:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	Atmosphere learning malfunction in A/F sensor circuit	×	1	A/F sensor	C	×
U0073:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	CAN system communication error (HS CAN)	—	—	Other	C, O, R	—
U0074:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	CAN system communication error (local CAN between PCM and TCM)	—	—	Other	C	—
U0101:00	ON	OFF	Flash	OFF	OFF	OFF	OFF	CAN communication: communication error to TCM	×	1	CCM	C	×
U0106:00	ON	OFF	OFF	OFF	OFF	OFF	OFF	LIN communication system: glow plug control module information communication error with PCM	×	1	CCM	C, O	×
U0121:00	ON	OFF	Flash	OFF	OFF	OFF	OFF	CAN communication: communication error to DSC HU/CM	×	1	CCM	C	×

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
U0131:00	OFF	OFF	Flash	OFF	OFF	OFF	OFF	CAN communication: communication error to EPS control module	—	1	Other	C	—
U0140:00	OFF	OFF	Flash	OFF	OFF	OFF	OFF	CAN communication: communication error to body control module (BCM)	—	1	Other	C	—
U0151:00	OFF	OFF	Flash	OFF	OFF	OFF	OFF	CAN communication: communication error to SAS control module	—	1	Other	C	—
U0155:00	ON	OFF	Flash	OFF	OFF	OFF	OFF	CAN communication: communication error to instrument cluster	×	1	CCM	C	×
U0214:00	OFF	OFF	Flash	OFF	OFF	OFF	OFF	CAN communication: communication error to start stop unit	—	1	Other	C	—
U0235:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	CAN communication: communication error to laser sensor	—	1	Other	C	—
U0298:00	OFF	OFF	Flash	OFF	OFF	OFF	OFF	CAN/LIN communication system: DC-DC converter information communication error with body control module (BCM) (without i-ELOOP) CAN communication: communication error to DC-DC converter (i-ELOOP) (with i-ELOOP)	×	1	Other	C	—
U0302:00 *3	ON	OFF	OFF	OFF	OFF	OFF	OFF	TCM processor error	×	1	CCM	C	×
U0315:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	DSC HU/CM error	—	1	Other	C	—
U0320:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	EPS control module error	—	1	Other	C	—
U0323:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Instrument cluster error	—	1	Other	C	—

DTC No.	Check engine light	Master warning light	i-stop warning light (amber)	Charging system warning light	Engine oil warning light	Wrench indicator light	Diesel particulate filter indicator light	Condition	Fail-safe function	Drive cycle	Monitor item	Self test type *1	Memory function
U0336:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	SAS control module error	—	1	Other	C	—
U0338:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Start stop unit error	—	1	Other	C	—
U0433:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Abnormal message from laser sensor	×	1	Other	C	—
U0599:00	OFF	OFF	Flash	OFF	OFF	OFF	OFF	Invalid data received from DC-DC converter	×	1	Other	C	—
U1007:00	OFF	ON	Flash	ON	OFF	OFF	OFF	CAN/LIN communication system: current sensor information communication error with body control module (BCM)	—	—	Other	C	—
U1100:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	LIN communication: communication error to engine oil level sensor	—	1	Other	C	—
U2300:00	OFF	OFF	OFF	OFF	OFF	OFF	OFF	Global central configuration error	—	1	Other	C	—
U3000:41	OFF	OFF	OFF	OFF	OFF	OFF	OFF	PCM processor error	—	1	Other	C	—

*1 : C: CMDTC self test, O: KOEO self test, R: KOER self test
*2 : MTX
*3 : ATX
*4 : Displays a message related to a charge system malfunction in the display. (With center display)
*5 : Glow indicator light flashes

Sending Continuous Monitoring System Test Results (pending code) (Mode 07)
- These appear when a problem is detected in a monitored system.

1-drive cycle type
- If any problems are detected in the first drive cycle, pending codes will be stored in the PCM memory, as well as DTCs.
- After pending codes are stored, if the PCM determines that the system is normal in any future drive cycle, the PCM deletes the pending codes.

2-drive cycle type
- The code for a failed system is stored in the PCM memory in the first drive cycle. If the PCM determines that the system returned to normal or the problem was mistakenly detected, and deletes the pending code. If the problem is found in the second drive cycle too, the PCM determines that the system has failed, and stores the pending codes, and the DTCs.
- After pending codes are stored, if the PCM determines that the system is normal in any future drive cycle, the PCM deletes the pending codes.